SimpleGrid plan recognition by grid size and observations

						max-term			sharpsat-td			mc-sat		
Grid size	Goals	Min goal distance	Obser-vations	Mean CNF variables	Mean CNF clauses	Mean time (s)	Among top	Is top	Mean time (s)	Among top	Is top	Mean time (s)	Among top	Is top
3×3	2	3	1	141	819	11.7	8/8	0/8	15.7	5/8	3/8	14.6	4/8	4/8
3×3	2	3	2	141	820	12.1	8/8	0/8	16.0	4/8	4/8	14.2	4/8	4/8
4×4	3	5	1	400	2,610	16.2	12/12	0/12	20.1	8/12	0/12	33.4	6/12	6/12
4×4	3	5	2	400	2,611	16.2	12/12	0/12	20.1	8/12	4/12	32.9	4/12	4/12
4×4	3	5	3	400	2,612	15.8	12/12	0/12	19.8	4/12	4/12	32.3	5/12	5/12
5×5	4	6	1	812	5,622	25.6	16/16	0/16	35.0	6/16	2/16	219.4	6/16	6/16
5×5	4	6	2	812	5,623	34.4	16/16	0/16	36.0	7/16	3/16	213.4	2/16	2/16
5×5	4	6	3	812	5,624	43.9	16/16	0/16	35.0	7/16	5/16	207.4	3/16	3/16
5×5	4	6	4	812	5,625	36.1	16/16	0/16	34.6	4/16	4/16	200.1	5/16	5/16
6×6	5	7	1	1,557	11,192	31.2	20/20	0/20	37.7	4/20	4/20	350.0	4/20	4/20
6×6	5	7	2	1,557	11,193	28.8	20/20	0/20	37.7	5/20	3/20	350.1	3/20	3/20
6×6	5	7	3	1,557	11,194	28.8	20/20	0/20	37.1	4/20	4/20	349.9	3/20	3/20
6×6	5	7	4	1,557	11,195	29.7	20/20	0/20	37.9	6/20	4/20	349.6	5/20	5/20
6×6	5	7	5	1,557	11,196	29.6	20/20	0/20	37.7	4/20	4/20	346.2	2/20	2/20
7×7	6	8	1	2,436	17,937	40.5	24/24	0/24	149.4	3/24	3/24	738.8	4/24	4/24
7×7	6	8	2	2,436	17,938	40.3	24/24	0/24	100.8	5/24	5/24	739.4	3/24	3/24
7×7	6	8	3	2,436	17,939	40.2	24/24	0/24	95.9	5/24	5/24	738.5	4/24	4/24
7×7	6	8	4	2,436	17,940	39.8	24/24	0/24	95.6	7/24	7/24	737.3	6/24	6/24
7×7	6	8	5	2,436	17,941	39.8	24/24	0/24	94.7	7/24	7/24	734.3	8/24	8/24
7×7	6	8	6	2,436	17,942	39.6	24/24	0/24	94.6	4/24	4/24	735.1	3/24	3/24
8×8	7	9	1	4,024	30,206	61.3	28/28	0/28	–	0/5 (5 run)	0/5 (5 run)	–	0/5 (5 run)	0/5 (5 run)
8×8	7	9	2	4,024	30,207	61.3	28/28	0/28	–	0/5 (5 run)	0/5 (5 run)	–	0/5 (5 run)	0/5 (5 run)
8×8	7	9	3	4,024	30,208	61.2	28/28	0/28	–	0/5 (5 run)	0/5 (5 run)	–	0/5 (5 run)	0/5 (5 run)
8×8	7	9	4	4,024	30,209	61.1	28/28	0/28	–	0/5 (5 run)	0/5 (5 run)	–	0/5 (5 run)	0/5 (5 run)
8×8	7	9	5	4,024	30,210	61.6	28/28	0/28	–	0/5 (5 run)	0/5 (5 run)	–	0/4 (4 run)	0/4 (4 run)
8×8	7	9	6	4,024	30,211	61.1	28/28	0/28	–	0/5 (5 run)	0/5 (5 run)	–	0/4 (4 run)	0/4 (4 run)
8×8	7	9	7	4,024	30,212	61.0	28/28	0/28	–	0/4 (4 run)	0/4 (4 run)	–	0/4 (4 run)	0/4 (4 run)

Fully connected grids with 1 truck and multiple possible goal destinations. “Among top”: the runs in which the true goal has the highest posterior, tied or alone. “Is top”: the runs in which the true goal alone has the highest posterior. Examples (evidence files): 3×3 16, 4×4 36, 5×5 64, 6×6 100, 7×7 144, 8×8 196. Time limit per run: 1 minute (3×3, 4×4); 15 minutes (5×5, 6×6); 20 minutes (7×7, 8×8). Runs that did not finish (counted as misses, left out of the mean times; the per-size tables say why): 7×7 sharpsat-td 36, mc-sat 2; 8×8 sharpsat-td 34, mc-sat 32. Not run (the method was stopped at that size; its table says how many of the runs made finished): 8×8 sharpsat-td: 162; 8×8 mc-sat: 164.